Journal of Petroleum Science and Engineering 175 (2019) 375–388

Contents lists available at ScienceDirect

Journal of Petroleum Science and Engineering

journal homepage: www.elsevier.com/locate/petrol

Effect of temperature on nano- and microscale creep properties of organic-rich shales

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ARTICLE INFO

Keywords:
Organic-rich shales
Short-duration nanoindentation
Creep nanoindentation
Creep rate magnitude
Clay-bound water
High temperature

ABSTRACT

Understanding the creep properties of gas shales at elevated temperatures is critical for accurately predicting reservoir performance and assessing the effect of viscous behavior on closure rate, production rate, and conductivity loss. In order to better understand creep properties at in-situ conditions, a quantitative relationship between temperature, mechanical properties, and complex chemo-physical mechanisms that cause thermal alteration needs to be obtained. In this study, the elastic, strength, and creep properties of organic-rich shales are investigated by a combination of nanoindentation, energy dispersive x-ray spectroscopy, and micromechanical modeling over a range of temperatures (23–350 °C). Within the allotted temperature range, the elastic and strength properties of the porous clay/kerogen phase remain unchanged. A similar behavior is noticed in the creep modulus from temperatures 23–200 °C where it remains relatively unchanged and shows an isotropic characteristic. However, the creep modulus of the clay phase increases at 300 °C in both parallel and perpendicular directions to the bedding plane. While the analysis of creep data at lower temperatures shed light on the dictating role of organic matter and porosity on creep properties of organic-rich shales (Slim et al., 2018), at high temperature (above 200 °C) creep properties are affected by removal of clay-bound water. The results show that this small water content assists in mechanical compaction and grain sliding responsible for time-dependent deformation. We argue that the removal of clay-bound water at temperatures above 200 °C increases the frictional coefficient between clay particles and between clay particles and organic matter.

1. Introduction

The relatively low permeability of organic-rich source rocks makes hydraulic fracturing an essential process for the production of hydrocarbons from such formations. Several factors cause the loss of fracture conductivity, which results in low recovery and inefficient production. One of the major contributors of the drop in fracture conductivity is the creep of the rock (i.e. deformation over time under constant stress) not captured by linear elasticity. Thus, failure to characterize the time-dependent behavior of these source rocks may result in errors in estimating reservoir performance during depletion.

Numerous experimental and theoretical studies have dealt with obtaining creep properties of organic-rich source rocks under ambient conditions (Li and Ghassemi, 2012; Sone and Zoback, 2013; Rassouli and Zoback, 2016). However, these conditions do not capture the in-situ behavior of many source rocks. Moreover, recent progress in thermal recovery of shale gas requires an enhanced understanding of reservoir properties at elevated temperatures. Therefore, a quantitative relation between temperature, mechanical properties, and complex chemo-physical mechanisms that cause the thermal alteration needs to be obtained.

Creep of organic-rich shales has mostly been investigated at the macroscale of engineering applications (Li and Ghassemi, 2012; Sone and Zoback, 2013, 2014). However, the physical source of creep in source rocks, the contribution of individual constituents, and the controlling factors are still poorly understood. Considering the characteristic size of different material phases in organic-rich shales (e.g. clay, calcite, and quartz), the assessment of their mechanical properties requires the use of nanoscale testing methods, such as nanoindentation. Nanoindentation has been proven to be a well-suited technique in measuring the mechanical properties of organic-rich shales (Abedi et al., 2016a; Slim et al., 2018; Slim, 2016). Moreover, recent developments have enabled the ability to perform instrumented nanoindentation at elevated temperatures with the use of temperature control stages. High temperature nanoindentation testing has advanced considerably over the past decades and has been used to extract

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https://doi.org/10.1016/j.petrol.2018.12.039
Received 31 August 2018; Received in revised form 28 November 2018; Accepted 14 December 2018
Available online 21 December 2018
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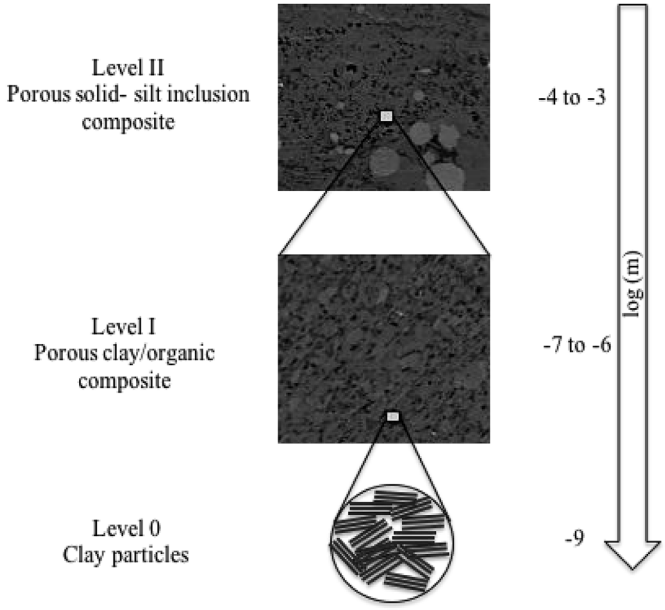


Fig. 1. Multiscale thought-model of organic-rich shale. Level 0 represents the scale of elementary clay particles at nanometer length scales. Level I corresponds to the scale of porous clay composite intermixed with organic matter at the micro and sub-micrometer length scales (scale of indentation), with the porosity representing the mesoporosity. Level II represents the level of porous organic/inorganic hard inclusion composite [after Abedi et al., 2016b].

temperature dependent mechanical properties of various types of materials (Beake and Smith, 2002; Schuh et al., 2006; Duan and Hodge, 2009; Trenkle et al., 2010; Everitt et al., 2011; Korte et al., 2011; Wheeler and Michler, 2013).

Shale rocks are complex naturally occurring geomaterials with multiscale heterogeneity and transverse isotropy. In this study, we follow the microstructural thought model in Fig. 1 for organic-rich shales, presented by Abedi et al. (2016b). In this three-level model, level 0 corresponds to the scale of clay particles. Level I is the scale of solid clay particles mixed with porosity and organic matter that form the clay/organic composite at the micro and sub-micrometer length scales. This scale corresponds to the scale of nanoindentation and some surface observational techniques such as SEM and EDX and is the scale of our study in this work. The particular significance of this level is due to the fact that significant creep deformation of shales is believed to be due in large part by porous clay/organic composite phase (Slim et al., 2018; Rassouli and Zoback, 2018; Sone and Zoback, 2014). Lastly, level 0 represents the scale of clay particles.

In this project, we investigate elastic, strength, and creep properties of porous clay/organic composite phases within an organic-rich shale sample by performing short-duration and creep nanoindentation tests at ambient and elevated temperatures. The results illustrate the mechanisms dictating creep properties of organic-rich shales and emphasize the role of clay-bound water in the time-dependent response. Such an understanding could lead to an improved prediction of hydrocarbon recovery and an optimized hydraulic fracturing operation for economic production.

The paper is organized as follows: The first section presents the studied material and the methods that have been used in order to study time-independent and time-dependent mechanical properties of porous clay/organic phases. The outcome is presented in the results and discussion section which reflects the effect of elevated temperature on the results from short-duration and creep nanoindentation tests.

Table 1
X-Ray Diffraction (XRD) and Mercury Injection Porosimetry (MICP) for Haynesville sample, and Rock Eval results for Haynesville shale.

Mineralogy	Haynesville Shale (wt. %)
Clay	27.5
Quartz	23.1
Carbonate	40.9
Pyrite	1.40
Feldspar	6.20
Apatite	0.90
Porosity (%)	3.20

2. Materials and methods

2.1. Materials

Shale sample from the Haynesville formation was used in this study. The sample contains both organic matter and inorganic minerals and was extracted from a gas-rich zone (Wang et al., 2017). The mineral composition and porosity of the sample were obtained by X-ray diffraction (XRD) and Mercury Injection Porosimetry (MICP), respectively (Table 1). Illite and Chlorite are the main clay components in the studied sample (Valdes and Heidari, 2017). Rock eval pyrolysis was performed to evaluate geochemical characterization an of organic matter before and after exposure to high temperature (300 °C) in order to assess any possible changes to organic matter due to heat exposure. All results are provided in Table 2. Tmax represents the temperature at which the maximum rate of hydrocarbon generation occurs during pyrolysis. The magnitude of Tmax obtained in both cases of before and after exposure to high temperature represent very mature kerogen. S1 indicates the amount of free hyrocarbons in the sample. S2 represents the amount of hydrocarbons generated during thermal pyrolysis, and S3 is indicative of CO_2 generated through thermal breakdown of organic matter. The difference in total organic carbon (TOC), Tmax, S2, and S3 between experiments performed before and after exposure to high temperature are due to very heterogeneous nature of shale rocks (due to crushing of samples for rock eval pyrolysis, it was not possible to test the same sample) and is in the range of experimental error. The results indicate that the highest temperate being used in this study for creep analysis (300 °C) does not impose any geochemical changes to organic matter present in the sample. The S1 peak is expected to decrease when the shale sample is exposed to high temperature due to release of free hyrocarbons as the sample is being heated.

Before conducting the experiments, shale rock samples are prepared and polished to have the smooth surfaces required for sensitive surface testing methods like nanoindentation and EDX. Shale specimens were cut from core plug by diamond saw into 0.5–1 cm thick and 1 cm diameter pieces. Surfaces were prepared in parallel to bedding (X1) and normal to bedding (X3) directions. Specimens were coarse polished first with 400 grit hard perforated pads and an oil-based diamond suspension in order to prevent chemical interactions. The next stage of polishing consists of drying polishing of specimens using 9-, 3-, and 1- and 0.3 μm aluminum oxide pads abrasive disks successively. Specimens are

Table 2
Rock Eval results for Haynesville samples before and after exposure to high temperature (300 °C).

	Before heating	After heating
Tmax (°C)	495	462
S1 (mg Hc/g rock)	0.79	0.16
S2 (mg Hc/g rock)	0.37	0.54
S3 (mg Hc/g rock)	0.30	0.39
TOC (%)	2.77	2.62

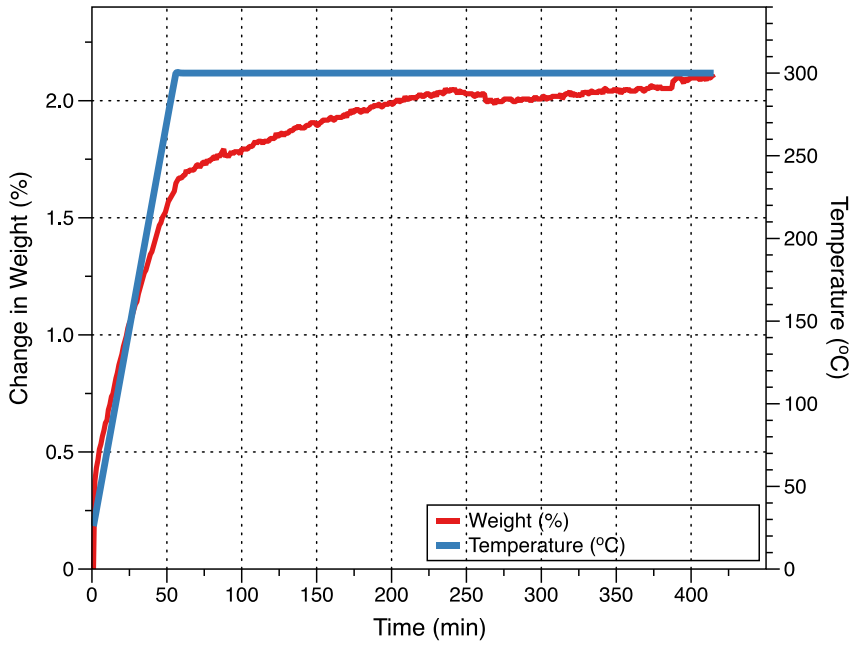


Fig. 2. Thermogravimetry Analysis (TGA) performed on Haynesville shale at 300 °C; figure shows the change in weight (%) (primary axis) and temperature (°C) (secondary axis) versus time (min) (x-axis).

ultrasonicated in n-decane solution in between polishing steps (Abedi et al., 2016a). The AFM measured root-mean-squared (RSM) roughness of shale samples prepared with the same procedure was about 30–150 nm (Bobko, 2008; Deirieh, 2011).

2.2. Thermogravimetry Analysis

The weight (%) change of the Haynesville sample as a function of temperature was measured by Thermogravimetry Analysis (TGA). The test was performed from room temperature (23 °C) to 300 °C at a heating rate of 5 °C/min. The temperature was held constant for 6 h at 300 °C to investigate the weight change of the sample during high temperature creep indentations. Fig. 2 shows the change in weight (%) and temperature (°C) as a function of time (min). The beginning portion of weigh loss accounts for the loss of surface and pore water. In a separate experiment, we hold the sample at 100 °C and 200 °C for 2 h to examine any weight change during indentation performed at these temperatures and the results showed negligible weight loss.

2.3. Assessment of mechanical properties by indentation

2.3.1. Elastic and strength properties from short-duration indentation tests

Grids of short-duration nanoindentation tests are performed on the surface of shale samples (Abedi et al., 2016a; Deirieh et al., 2012). At each point of the grid, an indenter tip (here a 3-sided pyramidal Berkovich tip) is orthogonally pushed onto the surface of the material; both the load applied to the tip and the penetration of the indenter tip into the sample are measured. The mechanical properties of the indented material, indentation modulus M and indentation hardness H, are then determined by employing continuum-based constitutive models to the force-displacement curve resulting from the experiment (Oliver and Pharr, 2004):

$$M = \frac{\sqrt{\pi}}{2}\frac{S}{\sqrt{A_c}} \tag{1}$$

$$H = \frac{p_{\max}}{A_c} \tag{2}$$

where $S = \frac{dP}{dh}$ is the initial slope of the unloading stage of the load-displacement (P-h) curve, A_c represents projected contact area between indented material and the indenter tip (Oliver and Pharr, 2004), and $P_{\max}$ is the maximum indentation load.

The obtained indentation modulus and hardness are associated with the elastic and strength properties of the material (Bobko et al., 2011; Cariou et al., 2008; Galin, 1953; Ganneau et al., 2006). Indentation modulus of an isotropic elastic medium is related to the plane-strain modulus $M = E/(1-\upsilon^2)$ in which E is the Young's modulus and υ Poisson's ratio of the material. For the case of a transversely isotropic material, which includes shale rocks, Delafargue and Ulm (2004) have provided an explicit approximation of the two indentation moduli in parallel to bedding and normal to bedding directions, M_1 and M_3, as a function of the five elastic constants C_{ij} of the material (Mashhadian et al., 2018a, 2018c).

Indentation hardness is often related to strength of the material. For the case of metals, Tabor (1948) suggested a simple relation between indentation hardness H and tensile strength σ_y : $H/\sigma_y \approx 3$. For the case of cohesive-frictional material, Bobko et al. (2011) provided a solution that allows one to determine the cohesion and friction of a material whose strength behavior follows the Drucker-Prager model based on the dependence of hardness to cohesion and friction coefficient.

In every indentation tests, a trapezoidal load history was applied by using a maximum load of $P_{\max}$ = 4.8 mN registered within 10s, detained constant over 10s, and released within 10s. Several indentation grids were performed on the surface of specimens. As a result of a massive indentation campaign, Bobko (2008) found out that the aforementioned experimental parameters solicit porous clay composite based on consideration of the roughness of surface and scale separation criteria (i.e., the characteristic size of the representative elementary volume must be greater than the size of the largest heterogeneity which is the size of pores and kerogen aggregates in this study). Finite element analysis of indentation experiment with Berkovich tip indicate that the probed volume in an indentation test is about 3–5 times the indentation depth. The average of the maximum indentation depth within the porous clay/organic phase is 480 nm parallel to bedding plane (X1) and 670 nm in normal to the bedding plane (X3) in this study. The depth of penetration increases by about 60 nm in both X1 and X3 direction during the holding phase (in nanoindentation creep experiments), thus making the average material volume indented within the porous clay/

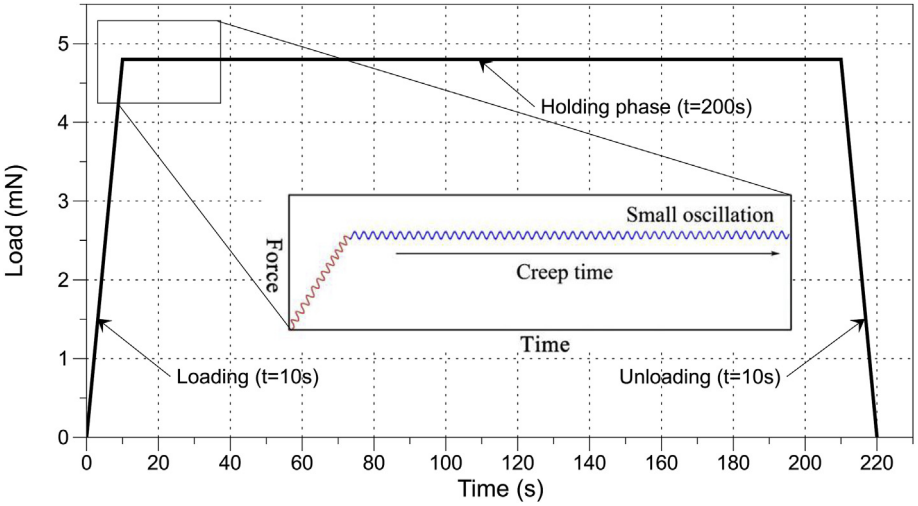


Fig. 3. Schematic representation of a load-time curve of the creep nanoindentation experiments used in this study. The insert represents the superimposed oscillatory force in nano-DMA testing protocol (insert modified after Bruker nano-DMA manual).

kerogen phase in the range of 1–3 μm. We chose grid spacing of $l = 7$ μm which provides enough separation between indents. Each grid consisted of 400–625 indents (20 × 20 or 25 × 25 grid) covering about 140 μm × 140 μm–175 μm × 175 μm of the sample surface which results in statistical independence and sufficient sampling representation. This size of grid also ensures adequate resolution in our EDX analysis (Section 2-5).

2.3.2. Creep rates from indentation tests

Compared to short-duration nanoindentation testing, the nanoindentation creep experiments consist of a relatively long holding phase (200s in this study, Fig. 3). Throughout the holding phase, penetration of the tip and consequently the contact area between the tip and the sample increases. The loading and unloading phases are fast enough (10s) to minimize the effect of plasticity during loading and viscous effects during loading and unloading (Vandamme and Ulm, 2013). Similar to the short-duration nanoindentation experiments, a grid indentation technique is used for the creep measurements due to the heterogeneous and multi-phase nature of organic-rich shales. Nano-dynamic mechanical analysis (nano-DMA), a testing protocol used to infer depth during lengthy creep tests, is used in this study (Hysitron, 2011). In this testing protocol, the quasistatic force is retained constant, while a small oscillation with a "reference frequency" is overlaid, to infer indenter-material contact area separately from the measured depth (Fig. 3). A 5 s reference section is added to the creep hold phase (200 s in our study) to acquire "reference tests parameters" (e.g., reference indentation modulus). This reference frequency technique enables thermal drift correction during the course of the experiment as well as continuous and precise measurement of mean contact pressure and indenter penetration depth over time. More information about nano-DMA testing protocol and associated parameter analysis is presented in Appendix A.

When a linear viscoelastic material is loaded over positive time ($t > 0$) in a triaxial creep experiment, the output strain response can be described through the integral form as follows:

$$\varepsilon(t) = \int_0^t J(t-\tau)\frac{d\sigma(\tau)}{d\tau}d\tau \tag{3}$$

where $J(t)$ is the creep compliance function that characterizes the time-dependent strain response to sudden loading. Similar to a triaxial creep test and depending on the magnitude of imposed stress, the creep indentation test is expected to describe the creep properties of the material by determining the contact creep compliance. The creep indentation experiment can be approximated by a Heaviside step loading in the form:

$$P(t) = P_{max}H(t) \tag{4}$$

where $P_{\max}$ is the maximum applied load and $H(t)$ is the Heaviside step function.

In the indentation experiment, the Galin-Sneddon solution for indentation of an isotropic solid relates the indentation load $P(t)$ to the indentation penetration depth $h(t)$ and its evolution over time by:

$$P(t) = \frac{2\tan\theta}{\pi}Mh^2(t) \tag{5}$$

where θ represents the half-angle of the equivalent conical indenter tip and M corresponds to the indentation modulus.

A well-established method for solving boundary value problems comprising viscoelastic materials is considered to be the correspondence principle of linear viscoelasticity. Based on this method, all time-dependent elastic field variables and constitutive properties can be replaced by their corresponding s-multiplied Laplace transform viscoelastic analogs, solved in the Laplace domain, and then converted back to the time domain. Lee and Radok (1960) showed that, for indentation problems, this correspondence principle can be applied as long as the contact area (or equivalently the penetration depth) increases monotonically. Applying the correspondence principle to Eq. (5) yields:

$$\overline{P(s)} = \frac{2\tan\theta}{\pi}s\overline{M(s)h^2(s)} = \frac{2\tan\theta}{\pi}\frac{\overline{h^2(s)}}{s\overline{L(s)}} \tag{6}$$

where the overbar represents the Laplace transformed quantity and s the Laplace parameter. $L(t)$ represents the contact creep compliance with $s\overline{L(s)} = \frac{1}{s\overline{M(s)}}$ (Vandamme et al., 2012; Vandamme and Ulm, 2013; Zhang et al., 2014). Vandamme and Ulm (2013) showed that the contact creep compliance rate $\dot{L}(t)$ can be obtained during the holding phase of the nanoindentation creep experiment from the following equation:

$$\dot{L}(t) = \frac{2a_u\dot{h}(t)}{P_{\max}} \tag{7}$$

where $\dot{h}(t)$ is the rate of penetration of the tip during the creep phase, $P_{\max}$ is the applied load throughout the creep phase, and a_u is the equivalent projected contact radius between the indentation tip and the indented material at the start of unloading (Slim et al., 2018). Eq. (7) was shown to be valid for an axisymmetric indenter probe and materials exhibiting time-independent plasticity (Vandamme et al., 2012). The contact creep compliance function, $L(t) - L(0)$ can be obtained by

integrating Eq. (7) with respect to time from the beginning of the holding phase (Zhang et al., 2014).

$$L(t) - L(0) = L(t) - \frac{1}{M} = \frac{2a_u \Delta h(t)}{P_{\max}} \tag{8}$$

In Eq. (8), t= 0 corresponds to the beginning of the holding time, where the contact creep compliance is equal to $L(0) = \frac{1}{M}$. Recent studies have shown that the contact creep compliance $L(t)$ is a material property, i.e., is a function that does not depend on the tip geometry and applied load magnitude used for the creep experiments (Vandamme et al., 2012). This contact creep compliance conveys as much information about the viscoelastic response of the material as the more common uniaxial creep compliance (Vandamme and Ulm, 2013). The contact creep compliance function can then be fitted with a logarithmic function of the form (Zhang et al., 2014):

$$L(t) - \frac{1}{M} = \frac{\ln\left(\frac{t}{\tau} + 1\right)}{C} \tag{9}$$

where C is the contact creep modulus and it has the same dimension as a modulus. The greater the contact creep modulus, the lower the creep rate of the material. The characteristic viscous time, τ specifies the time at which creep kinetics starts to show a logarithmic behavior. The rate of this logarithmic creep kinetics can be captured by the contact creep modulus:

$$\dot{L}(t) = \frac{1}{Ct}; \quad t \gg \tau \tag{10}$$

2.4. Indentation at elevated temperatures

In this study, the Hysitron® TI-950 nanoindenter equipped with a dual XSol temperature stage (max temperature of 800 °C) was used to perform indentations at elevated temperatures. The upper and lower temperature stages contain heating elements that maintain a uniform temperature throughout the sample (Fig. 4). The temperature stage comes equipped with a dry gas shroud connected to a dry gas source (argon) to prevent condensation on the sample when it is cooled to room temperature. The argon gas also helps reduce the occurrence of oxidation of the diamond tip and the sample, the latter being a key condition needed to replicate in-situ conditions.

The temperature of the top and bottom stages is measured in real time by the stage sensors and stored by the computer software. The stage can be programmed to heat the sample rapidly or at a constant slow rate. A plot of temperature versus time shows that allowing the sample to heat at a constant rate reduces the chance of exceeding the pre-set temperature value (Fig. 5). Fig. 5a shows the case when the temperature increases rapidly, at the beginning overshooting the set-point and then, after a while, stabilizing at the designated setpoint value. Fig. 5b represents the case when the temperature increases slowly and steadily to a setpoint with very slight overshoot. Moreover, since the goal of this investigation is to characterize the effect of exposure to high temperatures on mechanical properties of organic-rich shales, the heating procedure needs to be designed to limit the corresponding thermal gradients and transient effects of heating. Therefore, specimens were heated at a rate of 1 °C/min using the constant rate method, also known as the “ramp” feature.

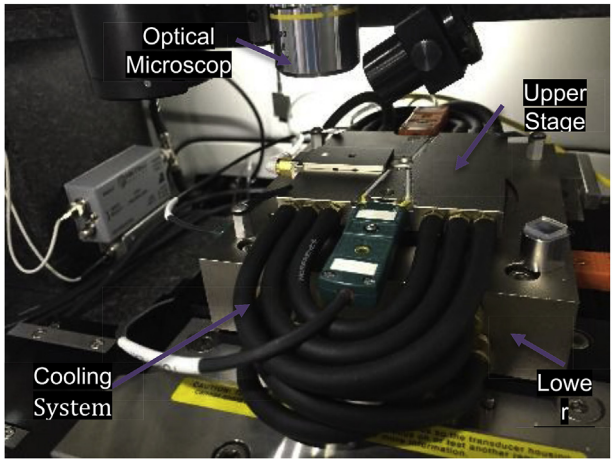


Fig. 4. A Hysitron TI950 Nanoindenter temperature stage assembly (upper stage and lower stage) connected to a cooling system used for all temperature measurements in this study.

Increasing the temperature increases the thermal drift, and thus, adequate time should be given for the system to reach thermal equilibrium. For this reason, after the heating phase, the system temperature was held constant for 4 h to reach thermal equilibrium before starting the indentation. During the heating and equilibrium steps, the tip is brought into contact with the sample and then retracted 0.05 μm to account for sample expansion. The purpose of keeping the tip close (0.05 μm) to the sample surface is for thermal equilibration between the tip and the sample.

2.5. Coupling with EDX and clustering analysis

Organic-rich shales are extremely heterogeneous sedimentary rocks composed mainly of clay particles, organic-matter, silt-size inclusions such as quartz and feldspar, and trace amounts of other minerals. For the purpose of correlating mechanical properties obtained by nanoindentation experiments to material composition of the indented volume, semi-quantitative elemental concentrations obtained by energy dispersive x-ray spectroscopy (EDX) are mapped over nanoindentation grids (Mashhadian et al., 2018b; Abedi et al., 2016a). Such semi-quantitative EDX maps represent auto-scaled X-ray elemental intensities over the pre-indented area. EDX technique utilizes X-Ray spectrum radiated from the incited shale sample as a consequence of interference of electron beams with the surface of the sample. The radiated X-rays are then categorized depending on their energy and provide detailed spatial resolutions of elemental intensities in the micrometer range depending on the sample density and voltage of electron beam. The EDX compositional maps thus obtained do not characterize quantitative confined geochemistry at indented points, rather the auto-scaled intensity quantities are interpreted as qualitative measures of a material phase. Employing this approach in characterizing different chemo-mechanical phases, and considering the distinctiveness of the chemical composition of each material phase in the studied Haynesville shale specimens, each identified phase can be uniquely connected to one configuration of auto-scaled elemental intensities on average (Abedi et al., 2016a).

Although all main elemental components important for the characterization of specimens are obtained over the pre-conducted indentation grid, just a few elemental intensities are needed for suitable chemo-mechanical phase recognition. For example, in order to identify clay-rich phases (porous clay/organic phases), only the Si and Al elemental intensities are required. All maps are obtained using a LYRA FIB-SEM, equipped with a standard EDX Microanalysis System, or an Electron Microprobe, paired with a Thermo EDX System.

Results from the nanoindenter and EDX are coupled through averaging elemental intensities obtained by EDX over a 2 μm diameter area around each indentation point (to make the lateral resolution of EDX comparable to nanoindentation) and then by isolating different material phases using a multivariate clustering analysis approach (Fraley and Raftery, 2007; Abedi et al., 2016a; Mashhadian et al., 2018b). This statistical method is commonly employed to identify the optimum number of Gaussian mixture distributions in a data set and the associated uncertainties.

Adequate comparison of the representative length scales of the

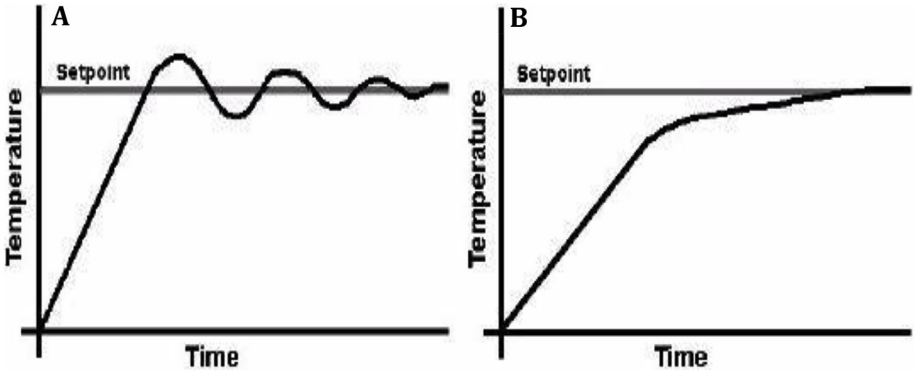


Fig. 5. Increasing the stage temperature before (left, 2A) and after (right, 2B) selecting the "ramp" feature [modified after Hysitron Temperature Stage Controls manual (2014)].

interaction volume is an essential aspect in coupling EDX and nanoindentation techniques (Abedi et al., 2016a). The length scale of the probed volume in EDX is ruled by accelerating voltage. Employing a 15 kV beam voltage, the interaction volume is about 2–3 μm for shale rocks (Abedi et al., 2016a; Deirieh et al., 2012). Moreover, as mentioned in Sec. 2-3-1, the size of the probed volume in an indentation test is about 3–5 times the indentation depth. With the average of the maximum indentation depth within the porous clay/organic phase being 480 nm parallel to bedding plane (X1) and 670 nm in normal to the bedding plane (X3) and with 60 nm increase in depth in both X1 and X3 direction during the holding phase (in nanoindentation creep experiments), comparable material volume is probed by each method.

To study the effect of temperature on the mechanical properties (elasticity, strength, and creep properties) of clay-rich phases (which are composed of clay aggregates and organic matter), the porous clay/organic phases are isolated by the above-mentioned method. Analysis was executed using R programming language with a multi-variate clustering package (mclust). Interested readers are referred to Abedi et al. (2016a) for more detailed discussion about the usage of clustering analysis in the identification of clay-rich phases.

Fig. 6 shows SEM images of Haynesville shale obtained at two magnification levels displaying the dispersed kerogen between clay particles in the porous clay/organic phase (Abedi et al., 2016a). Fig. 7 displays typical clustering analysis results obtained in one of our creep nanoindentation grids. Mechanical data (*M*, *H*, and *C*) and chemical data (auto-scaled elemental intensities of Si, and Al) are incorporated into the clustering analysis for suitable chemo-mechanical phase identification. Fig. 7b displays the mean mechanical properties of each phase. Phase 1 is identified as clay/kerogen phase (porous clay intermixed with kerogen) whereas phases 2 to 5 are identified as composite phase of clay and calcite, calcite, composite phase of calcite and quartz, and quartz phase respectively. Results show that the creep rate of isolated clay composite phase (phase 1) is much higher than the other phases (lower creep modulus, *C*) which is the case we have observed in every creep indentation experiment and is consistent with creep indentation results performed on other shale samples (Slim et al., 2018).

3. Results and discussion

3.1. Effect of temperature on elasticity and strength properties: short-duration nanoindentation

Short-duration nanoindentation grids were created in orthogonal directions on separately prepared samples for 23, 70, 250, and 350 °C. All the samples came from the same core and the spacing between them is in the range of few centimeters. In each grid, the porous clay/kerogen phases were identified employing the coupled EDX-nanoindentation clustering method and by including elemental intensities of Al and Si (as well as Ca or K depending on the mineralogy of the grid) along with mechanical data, indentation modulus, *M* and hardness, *H*.

Figs. 8 and 9 show a summary of the indentation results obtained in the porous clay/kerogen phases as a function of temperature. The results show that the stiffness and hardness of porous clay/kerogen phases remain approximately constant over the range of temperatures considered here. The contribution of clay and kerogen on indentation results obtained in porous clay/kerogen phase depends on their relative volume fractions in the indented volume. We can convert information

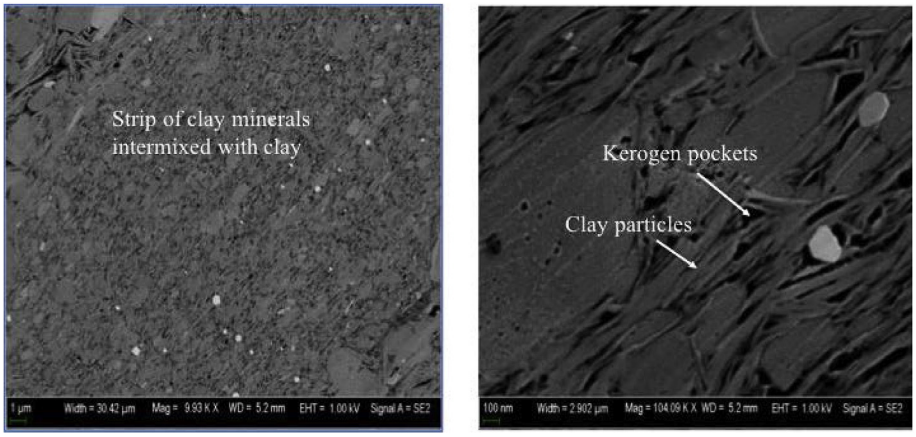


Fig. 6. SEM images of Haynesville shale obtained at two different magnification levels displaying the distribution of kerogen between clay particles in the porous organic-rich clay phase (adapted from Abedi et al., 2016a).

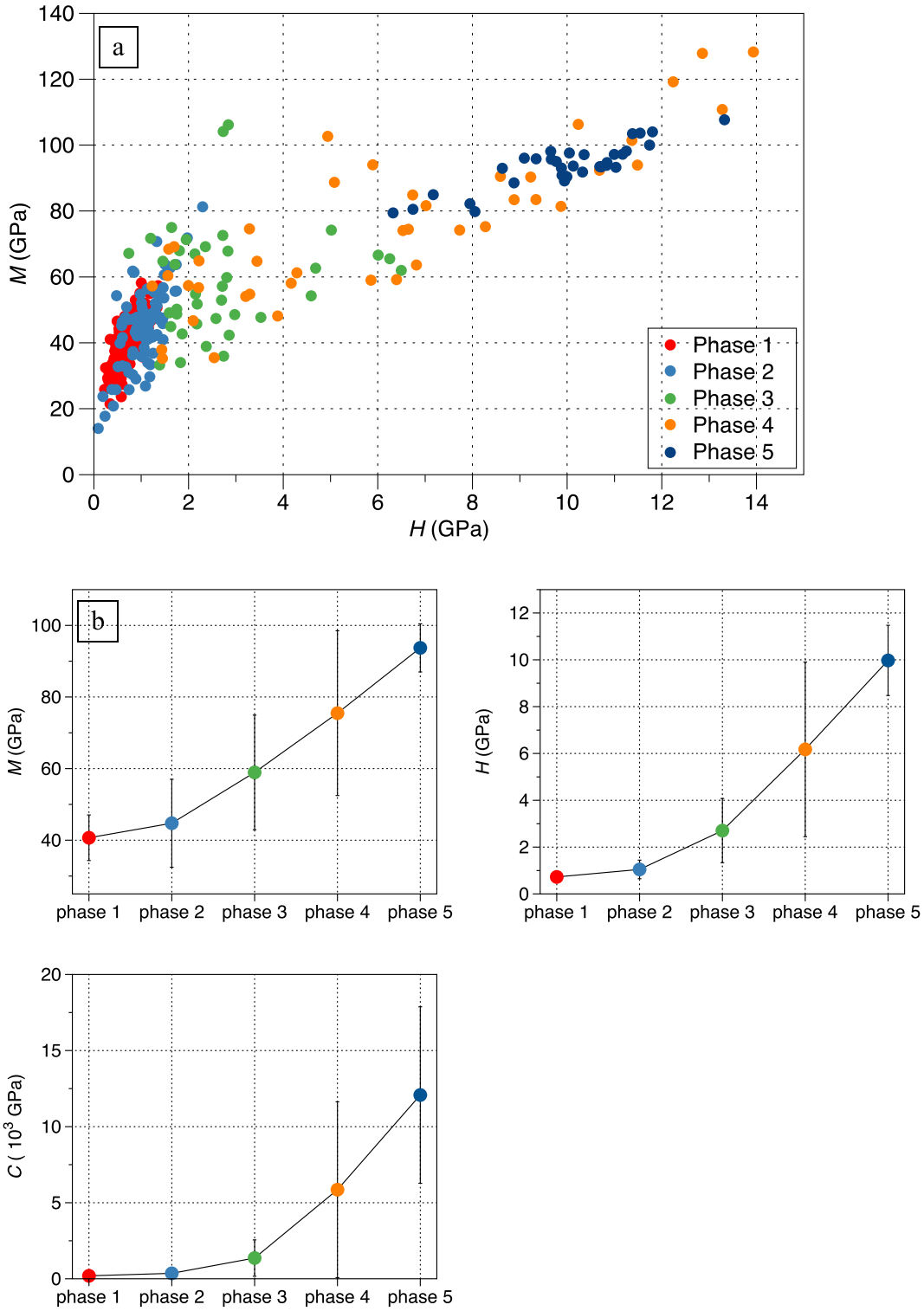


Fig. 7. Typical clustering analysis results obtained in one of our creep nanoindentation grids. Mechanical data (*M*, *H*, and *C*) and chemical data (auto-scaled elemental intensities of Si, and Al) are input values of the clustering analysis. Fig. 7b displays the mean mechanical properties of each phase.

regarding mineralogy and porosity of the sample into corresponding volumetric parameters (volume fraction of different material phases) at different levels of multi scale thought-model based on density of rock's constituents. Details of this type of analysis is explained in our previous publications extensively (Abedi et al., 2016a, 2016b) and a brief summary is presented here. Based on the studies performed by Fitzgerald et al. (1989), Kuila et al. (2014), and Vernik and Nur (1992), kerogen is mostly concentrated in the clay phase. Building upon the multiscale model proposed in Fig. 1, we can assume than at the microscopic level (nanoindentation level, level I, Fig. 1), the indented volume is comprised of clay particles, kerogen, and pore spaces: $\eta_c + \eta_k + \varphi = 1$, where η_c and η_k are volume fractions of clay and kerogen at microscopic level respectively and φ is the porosity within the porous organic-rich clay phase. Considering density of kerogen and clay to be 1.3 g/cm^3 and 2.75 g/cm^3 respectively, and porosity to be distributed evenly among all material phases, volume fraction of kerogen at level I

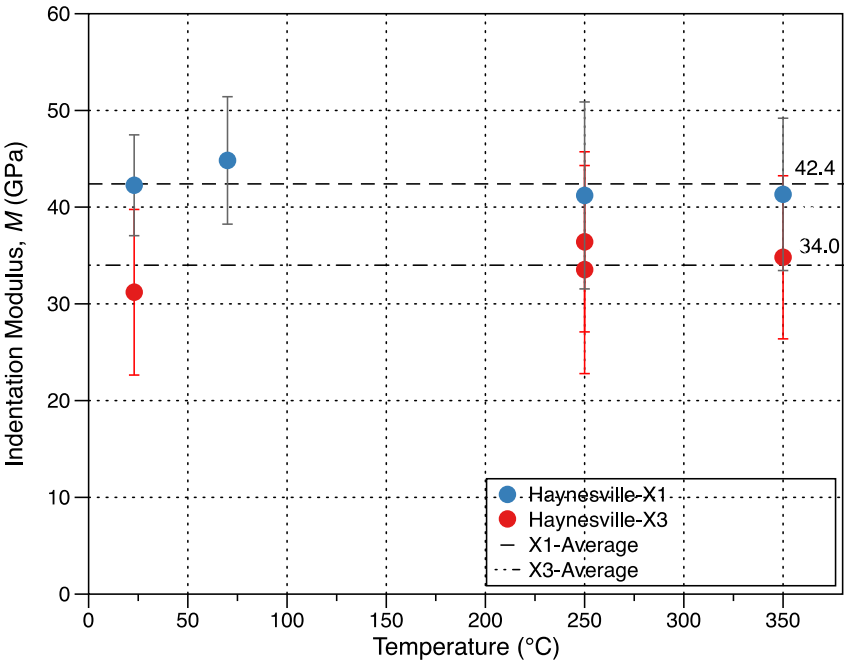


Fig. 8. Indentation modulus of clay/kerogen phase obtained from short-duration experiments for Haynesville shale samples parallel and perpendicular to bedding direction for a range of temperatures (23–350 °C).

(nanoindentation level) is calculated as 16.8% which is not negligible. Thus, variations in mechanical properties of kerogen due to exposure to high temperature would change indentation results. However, the effect would not be significant because of much higher volume fraction of clay compared to kerogen at nanoindentation level. Thus, the constant stiffness and hardness of porous clay/kerogen phases before and after exposure to high temperature may indicate toward the insensitive nature of the time-independent mechanical properties of kerogen and clay in the temperature ranges investigated in this study (up to 350 °C). The insensitive nature of kerogen with respect to temperature was also observed by Emmanuel et al. (2016) in a new AFM study investigating elastic properties of kerogen in shales during heating from room temperature to 225 °C. Ibanez and Kronenberg (1993) also observed weak temperature dependencies of illite-rich shales between temperatures of 22–196 °C and studies performed by Jarad et al. (2017) showed a negligible variation in effective stress-strain response of compacted illitic clay up to 69.2 °C. The results of rock eval pyrolysis performed on shale samples indicate no geochemical changes to organic matter present in the sample due to exposure to high temperature (Table 2).

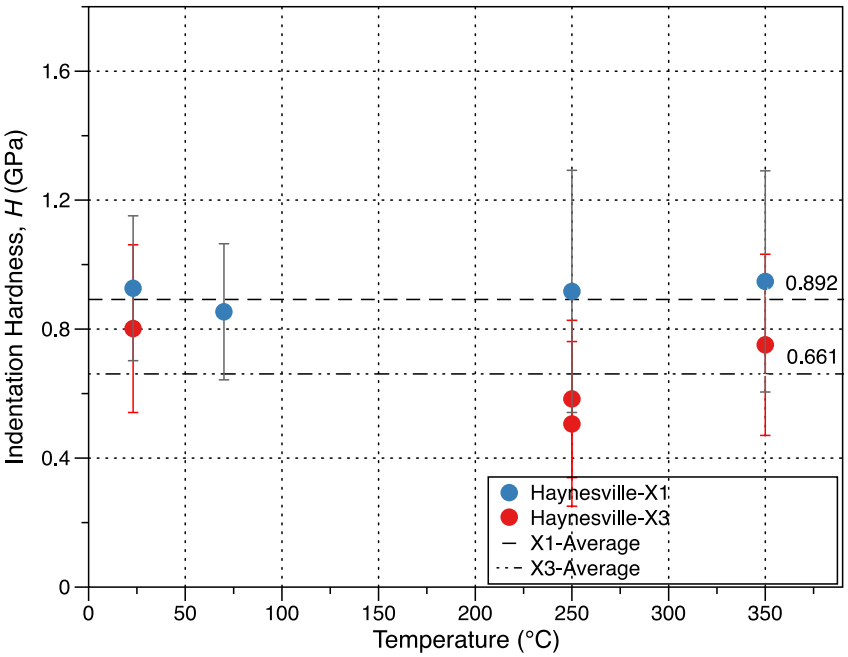


Fig. 9. Indentation hardness of clay/kerogen phase obtained from short-duration experiments for Haynesville shale samples parallel and perpendicular to bedding direction for a range of temperatures (23–350 °C).

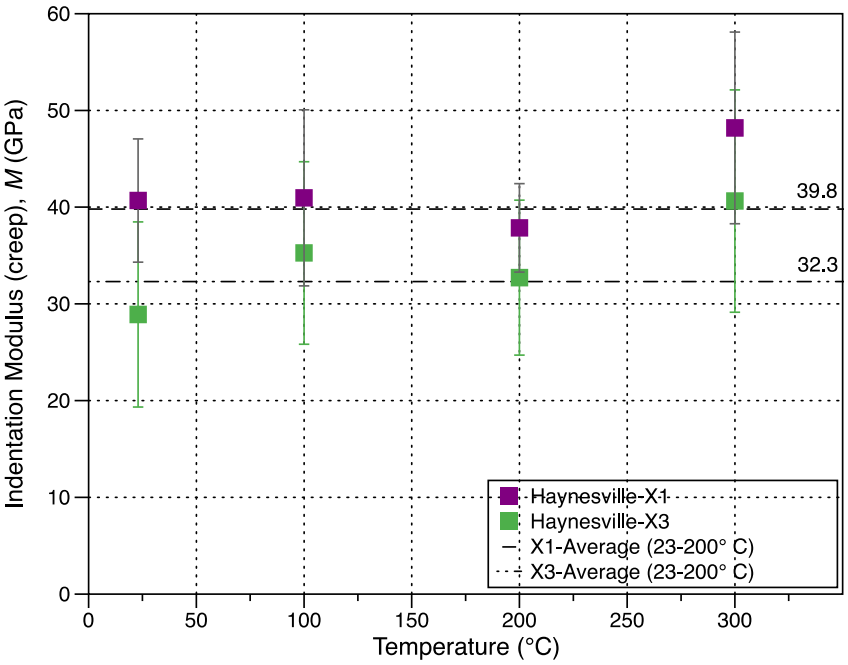


Fig. 10. Indentation modulus of clay/kerogen phase obtained from creep nanoindentation experiments for Haynesville shale samples parallel and perpendicular to bedding direction for a range of temperatures (23–300 °C).

3.2. Post creep elasticity and strength

Creep nanoindentation grids were performed in parallel to the bedding and normal to the bedding directions at 23, 100, 200, and 300 °C. Clustering analysis incorporating mechanical (*M*, *H*, and *C*) and chemical (Si and Al) data was performed to isolate the porous clay/kerogen phase in every creep nanoindentation grid. The indentation modulus and hardness obtained from the creep nanoindentation tests are presented in Figs. 10 and 11 as a function of temperature. Due to the longer holding phase of the creep experiments compared to the short-duration indentation tests, the penetration of the probe, and consequently the radius of the projected contact area, is higher in the creep experiments. This supports our reasoning behind the relatively smaller values of hardness in the creep experiments for temperatures 23–200 °C compared to obtained hardness values in standard (short-duration) indentation tests. On the other hand, the slight decrease in the indentation modulus from the creep experiments at 23–200 °C compared to the standard tests might be due to the changes in the packing density of the solid phase. At 300 °C, both the indentation hardness and modulus increased compared to the results obtained by short-duration indentation and the values at lower temperatures with a more significant increase in hardness (Fig. 12). These results suggest a

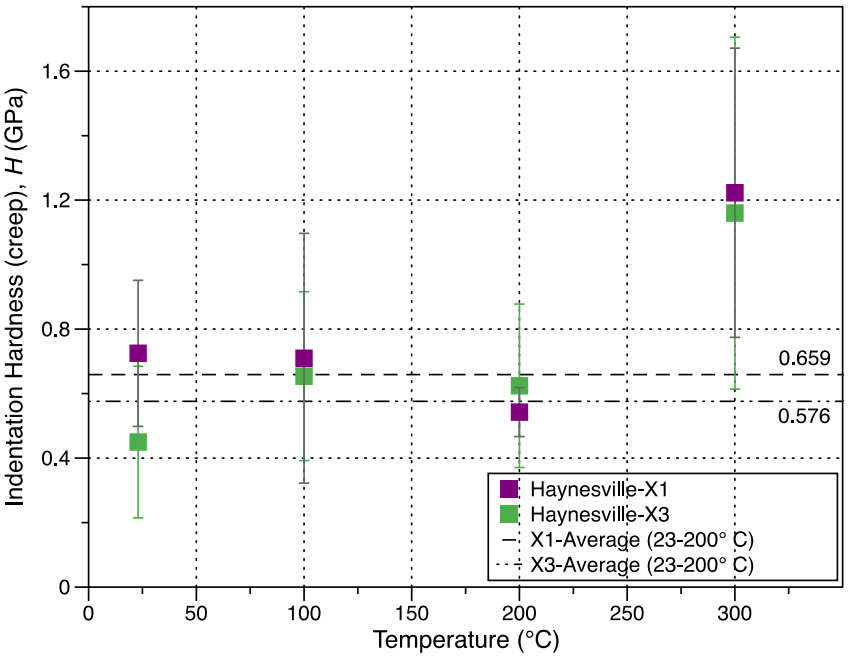


Fig. 11. Indentation hardness of clay/kerogen phase obtained from creep nanoindentation experiments for Haynesville shale samples parallel and perpendicular to bedding direction for a range of temperatures (23–300 °C).

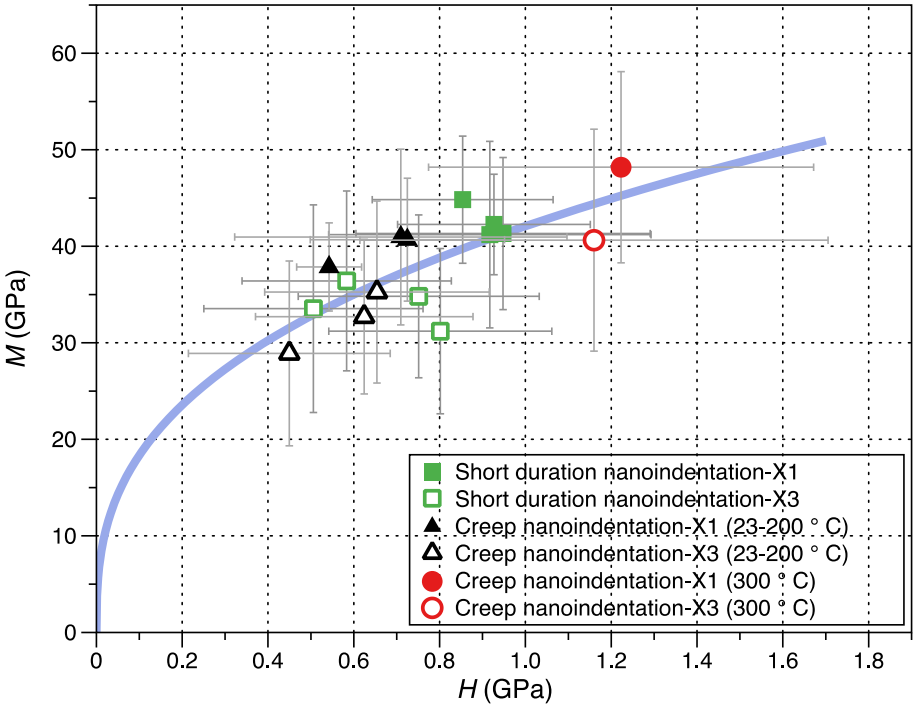


Fig. 12. A plot of indentation modulus, *M* versus indentation hardness, *H* obtained from performing short duration nanoindentation and creep nanoindentation tests for a range of temperatures.

stiffer material at the end of holding phase with less indent penetration ($H = \frac{P_{max}}{A_c(h)}$) and less compaction during the creep phase implying slower creep rate at 300 °C.

3.3. Effect of temperature on creep rates

Fig. 13 presents the contact creep compliance function obtained from a nanoindentation creep test performed on a few points within the porous clay/organic phase at temperatures 23 and 300 °C. Note that the observed trend is qualitatively identical to the trends seen in macroscopic triaxial creep experiments (Sone and Zoback, 2014), which indicates minutes-long nanoindentation creep tests can qualitatively capture the long-term creep kinetics of organic-rich shales (Zhang et al., 2014; Vandamme and Ulm, 2013; Slim et al., 2018; Slim, 2016). The contact creep modulus, *C*, can be obtained by fitting the contact creep compliance function with Eq. (9). The contact creep modulus

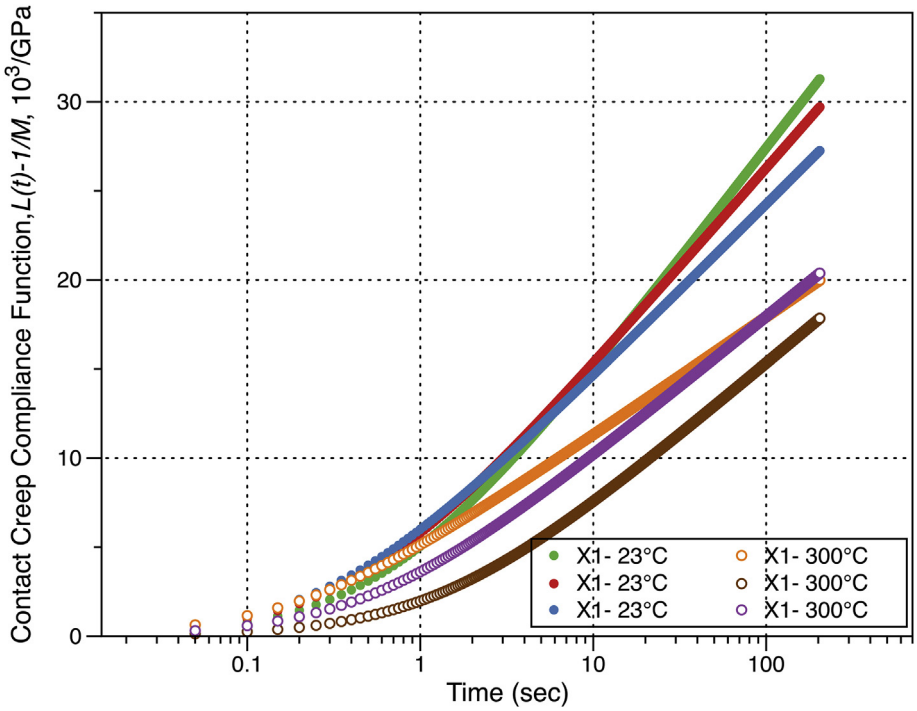


Fig. 13. Examples of six contact creep compliance functions obtained from creep nanoindentation experiments performed at 23 °C (three curves) and 300 °C (three curves) on a Haynesville shale sample, parallel to the bedding direction (X1).

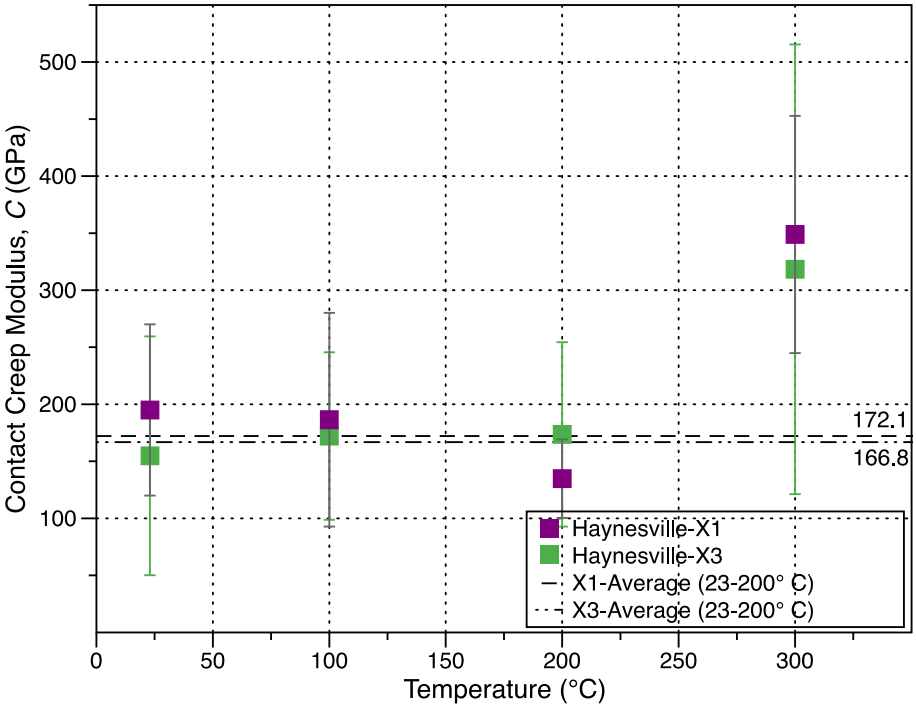


Fig. 14. Contact creep modulus data (y-axis) of clay/kerogen phase versus temperature (23–350 °C) for all creep experiments performed on Haynesville shale samples parallel and perpendicular to bedding direction.

characterizes the long-term creep properties of the probed material volume and represents its creep rate.

Fig. 14 shows the contact creep modulus of porous clay/kerogen phases as a function of temperature in orthogonal directions. Comparing the obtained creep moduli in parallel to the bedding and normal to the bedding directions, negligible anisotropy in creep rates in observed which indicates the prevailing role of isotropic phases, specifically organic matter and porosity, on creep properties of organic-rich shales. Moreover, the results show that the composite clay phase creeps slower at 300 °C in both parallel to bedding plane (X1) and normal to bedding plane (X3).

Considering the results obtained from Rock-eval pyrolysis, no significant changes in the amount of organic matter and maturity level due to exposure to high temperature (300 °C) have been observed (Table 2). Thus, evolution of organic matter is not responsible for the observed reduction in time-dependent mechanical properties. Crossplot of the contact creep modulus and creep indentation hardness (Fig. 15) show a linear scaling between *C* with *H* independent of the testing temperature. This result is in agreement with the recent study performed by Slim et al. (2018) in which they observed a maturity-dependent and scale-independent linear correlation between creep modulus and indentation hardness. This linear scaling between the two mechanical quantities (contact creep modulus, *C* and indentation hardness, *H*) indicate the fundamental creep deformation mechanism to be particle rearrangement and compaction within the clay composite phase (Rassouli and Zoback, 2018; Slim et al., 2018).

The weight loss of the sample exposed to high temperature (TGA results, Fig. 2) can be attributed to the release of free hydrocarbons and removal of clay-bound water at temperature above 200 °C. We argue that the removal of clay-bound water at high temperatures hinders grain sliding by increasing the frictional coefficient between clay platelets and between clay particles and organic matter (Rassouli and Zoback, 2018; Moore and Lockner, 2004). The effect of this small water content on grain sliding and creep deformation can be compared to the observed effect of biofilm formation on mechanical properties of sand particles. Perkins et al. (2000) observed that the presence of biofilm caused a negligible effect on the shear strength and stiffness of the Ottawa sand based results from on CU and CD triaxial tests, but that it boosted the creep deformation. The accelerated creep strain rate was linked to facilitating effect of biofilm on gradual reorientation of particles under long term stress (Perkins et al., 2000). Similar mechanism is happening in clay-rich phases during creep deformation. The presence of clay-bound water facilitates grain compaction and rearrangement and escalates creep deformation.

Although removal of clay-bound water decreases the creep deformation rate, the observed contact creep modulus is yet one order of magnitude smaller than that of muscovite (with a mean value of C = 5455; GPa, Slim et al., 2018). Similar to the role of clay-bound water on the creep rate, organic matter is reducing the frictional coefficient between clay particles and organic matter, causing the creep rate of porous clay/kerogen phase to be lower than that of clay particles. These observations signify the role of frictional resistance on driving viscous behavior of geomaterials.

Fig. 16 shows the experimental frequency density of the creep modulus for porous clay/kerogen at different temperatures in parallel to bedding and normal to bedding directions. Similar distribution of data is observed for creep modulus at 23–200 °C. However, the creep modulus of the porous clay/kerogen phase at 300 °C exhibits a shift and wider range that includes response from clay particles without the presence of bound water. We remind ourselves of the composition of the probed volume: porous clay (Illite/Chlorite) intermixed with kerogen. The wide range of nanoindentation creep results can be attributed to the varying quantities of different material phases (clay, kerogen and pore space) within the probed volume wherein at 300 °C includes the response of dry clay particles.

4. Conclusions

Short-duration and creep nanoindentation tests at ambient and elevated temperatures are performed to measure the elastic, strength, and creep properties of the porous clay/kerogen phase present within an organic-rich shale sample. We found that negligible change occurs in the time-independent properties (elastic and strength) of the porous clay/kerogen phase between temperatures 23–350 °C. This observation

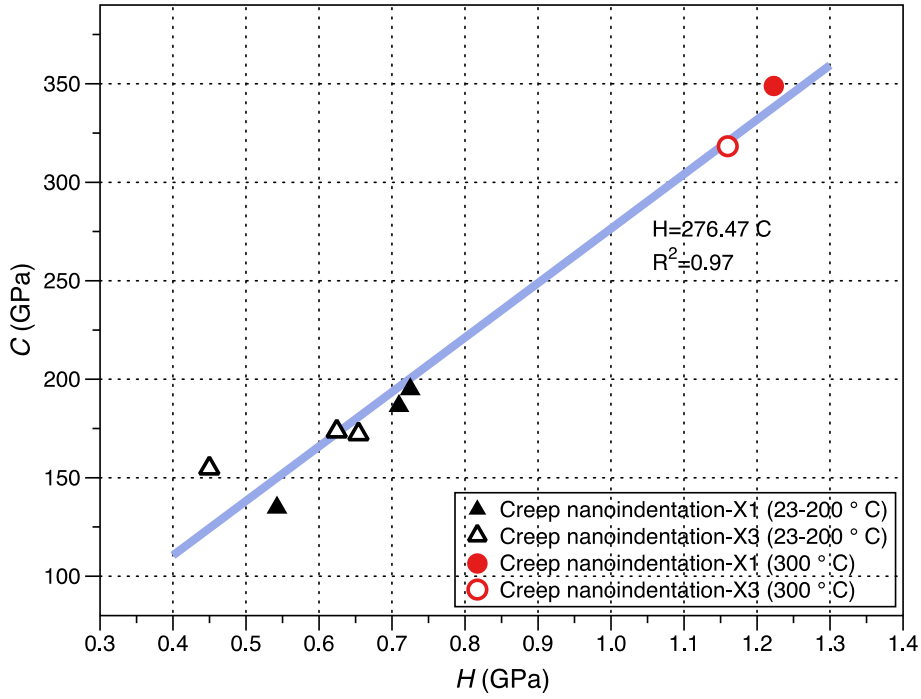


Fig. 15. Plot of contact creep modulus, C versus indentation hardness, H exhibiting a linear relationship between the two mechanical properties.

is consistent with the recent experimental results showing insensitive nature of time-independent mechanical properties of kerogen and illite-rich shales with respect to temperature (Emmanuel et al., 2016; Ibanez and Kronenberg,1993). The creep nanoinentation results over ambient and elevated temperature shows negligible anisotropy in creep rates when comparing creep modulus in parallel to bedding (X1) and normal to bedding (X3) directions, which indicates the prevailing role of isotropic phases, specifically organic matter and porosity, in driving creep properties of organic-rich shales. The results show a decrease in the creep rate (increased contact creep modulus) at 300 °C for both parallel and perpendicular to bedding plane, which we argue is a result of removal of clay-bound water at temperature above 200 °C. The removal of clay-bound water at high temperatures hampers grain sliding by increasing the frictional coefficient between clay platelets and between clay particles and organic matter. The post creep mechanical responses show an increase in both modulus and hardness of porous clay/organic phase at 300 °C with a more significant increase in hardness, an observation which again indicates toward the lower compaction rate (creep rate) at higher temperature. A closer examination of the creep results shows a shift and wider range of frequency density of the creep modulus at 300 °C that includes response from clay particles without the presence of bound water.

While our experiments shed light on the particle rearrangement and compaction as the fundamental creep deformation mechanism and the role of clay-bound water on grain sliding and creep deformation in shale rocks, our study focuses on a single shale formation. Further studies are required on both mature and immature shale samples to capture the time-dependent relationship present between organic matter (kerogen/bitumen), clay particals, and porosity from different depositional environments.

Acknowledgments

The authors acknowledge funding from the Harold Vance Department of Petroleum Engineering, Texas A&M University. The work was carried out in part in the Material Characterization Facility (MCF) at Texas A&M University.

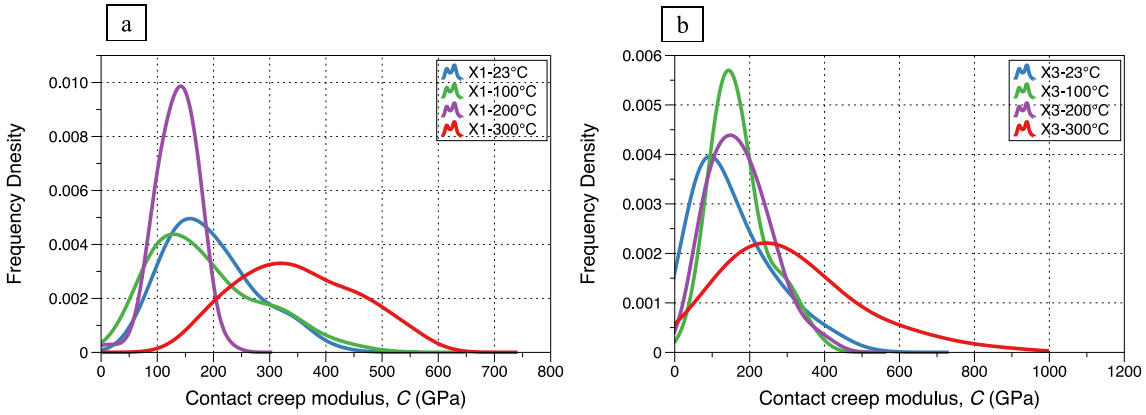


Fig. 16. Frequency density of contact creep modulus values (x-axis) for porous clay/kerogen phase indentations performed (a) parallel to bedding direction (X1) and (b) normal to bedding direction (X3).

Appendix A. Nano-dynamic mechanical analysis (nano-DMA)

Nano-dynamic mechanical analysis (nano-DMA), a testing protocol used to infer depth during lengthy creep tests, is used in this study (Hysitron, 2011). In this testing protocol, quasi-static force is retained constant, while a small oscillation with a "reference frequency" is overlaid, to infer indenter-material contact area separately from the measured depth (Fig. 2). A 5 s reference section is added to the creep hold phase (200 s in our study) to acquire "reference tests parameters" (e.g., reference indentation modulus). Throughout the reference section, the indenter oscillates at the reference frequency and the depth of the indenter is assessed, then the contact area is calculated, and the reference sample stiffness at the specific frequency K_{ref}, is quantified. The reference storage modulus, E'_{ref} is then obtained from the reference sample stiffness and the measured contact area during reference 5 s reference section (Hysitron, 2011):

$$E'_{ref} = \frac{K_{ref}\sqrt{\pi}}{\sqrt[2]{A(h_c)}} \tag{A.1}$$

where E'_{ref} is utilized for the remaining segment of the creep phase to obtain the contact area, A_c which requires knowledge of contact stiffness S:

$$A_c = \left(\frac{S\sqrt{\pi}}{2E'_{ref}}\right)^2 \tag{A.2}$$

In the dynamic analysis employed by Hysitron, an oscillatory motion is added by applying a small regulated force, having a frequency of ω and magnitude P_0 (Fischer-Cripps, 2011). Commonly, there is a phase difference, φ between the force and displacement response (Fischer-Cripps, 2011):

$$P(t) = P_0 e^{i(\omega t+\phi)} \tag{A.3}$$

$$h(t) = h_0 e^{i(\omega t)} \tag{A.4}$$

Considering a viscoelastic model shown in Figure A.1, the contact stiffness ($S = \frac{dP}{dh}$) can be obtained by combining the stiffness and damping coefficient (Fischer-Cripps, 2011):

$$\frac{P_0}{h_0} = \sqrt{(S + k_s - m\omega^2)^2 + \omega^2\lambda^2} \tag{A.5}$$

where k_s is the stiffness of the indenter shaft support spring, m is mass of the component obtained during air calibration, and λ is the damping coefficient. The phase angle between the force and displacement is determined from:

$$\tan\phi = \frac{\omega\lambda}{S + k_s - m\omega^2} \tag{A.6}$$

The transfer function, (A.5) depends on the type of viscoelastic model and the material-indenter interaction. For the case of a voigt model, the real and imaginary parts of the transfer function are (Fischer-Cripps, 2011):

$$TF_{re} = S + k_s - m\omega^2 \tag{A.7}$$

$$TF_{im} = \omega\lambda \tag{A.8}$$

where values of m and k_s can be obtained by fitting the transfer function to the results from nano-DMA air calibration. The application of oscillatory motion to the conventional quasi-static force enables a continuous measurement of S during the creep holding time. By accounting for the machine compliance, Equations A.5 and A.6 can be obtained from (Fischer-Cripps, 2011):

$$\frac{P_o}{h_o} = \sqrt{\left(\left(\frac{1}{S} + C_f\right) + k_s - m\omega^2\right)^2 + \omega^2\lambda^2} \tag{A.9}$$

$$\tan\phi = \frac{\omega\lambda}{\left(\frac{1}{S} + C_f\right)^{-1} + k_s - m\omega^2} \tag{A.10}$$

where C_f is considered as the compliance of the load frame.

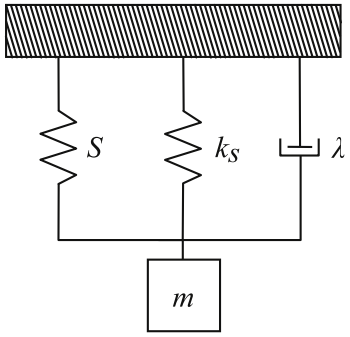


Fig. A1A representation of dynamic three-element model of nanoindentation instrument where S is the contact stiffness, λ is the damping coefficient, and k_s is the stiffness of the indenter shaft support springs (modified after Fischer-Cripps. 2011).

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